



IT-SIMPLERA INSTITUTE

Network Administration Internship and Training Program

Report Title:	Networking Fundamentals & Network Simulation Tools
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Tools Used:	GNS3, Cisco Packet Tracer, PuTTY

1. Introduction

This report documents the Week 01 assignment completed as part of the Network Administration Internship and Training Program at **IT-Simplera Institute**. The objective of this week was to build a foundational understanding of computer networking concepts and gain hands-on experience with two industry-standard network simulation tools: Cisco Packet Tracer and GNS3.

The assignment required the design and implementation of a simple Local Area Network (LAN) topology in both simulation platforms, followed by static IP address configuration and end-to-end connectivity verification using the ping command. All tasks were completed successfully before the submission deadline of 3 July 2026.

2. Networking Concepts Learned

Local Area Network (LAN)

A LAN connects devices within a limited area such as an office or building. In this assignment, a simple LAN was created using three PCs and one switch.

IP Addressing (IPv4)

- **Private IP Address:** A private IP address is used within a local network (such as a home or office) and cannot be accessed directly from the internet.
- **Public IP Address:** A public IP address is assigned by an Internet Service Provider (ISP) and is used to identify a device on the internet, making it accessible from outside the local network.

Every device on a network is assigned a unique IP address in the format of four octets (e.g., 192.168.1.1). In GNS3 the range 192.168.1.x was used; in Packet Tracer, 20.31.1.x was used.

Subnet Mask

A subnet mask defines the network and host portions of an IP address. The mask 255.255.255.0 (/24) allows up to 254 hosts on the same network.

Network Switch

A switch is a Layer 2 device connecting multiple devices within the same LAN. It forwards frames based on MAC addresses. Both the **GNS3** built-in switch and **Cisco 2960-24TT** were used.

Ping Command

Ping uses ICMP Echo Request and Reply messages to test connectivity. A successful ping with 0% packet loss confirms end-to-end communication.

VPCS (Virtual PC Simulator)

VPCS is a lightweight tool built into GNS3 that simulates end-user PCs, supporting basic IP configuration and ping commands.

3. Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a network simulation tool developed by Cisco Systems providing a visual drag-and-drop interface to design and simulate network topologies. Key interface areas include:

- Menu Bar: File, Edit, Options, View, Tools, Extensions, Help
- Logical/Physical Workspace: Main canvas where topologies are built
- Device Type Selection Box: Bottom left panel with routers, switches, PCs
- Specific Device Selection: Choose exact model (e.g., 2960-24TT switch)
- Realtime / Simulation Mode Toggle: Live mode or step-by-step packet tracing
- Connections Panel: Select cable type (straight-through, crossover, serial)

4. GNS3 Interface Overview

GNS3 (Graphical Network Simulator 3) is an open-source network emulator that runs real Cisco IOS images, making it closer to real-world networking than Packet Tracer. Key interface areas include:

- Toolbar: Start, Stop, Pause, reload devices; Add links; Zoom; Screenshot
- Device Panel (Left Sidebar): Browse routers, switches, end devices, and links
- Topology Canvas: Central workspace to drag, drop, and connect network devices
- Topology Summary (Right Panel): Lists all nodes and console port numbers

- Servers Summary: Shows local GNS3 server connection status (green = connected)
- Console Panel (Bottom): Displays GNS3 management logs, errors, and output

5. GNS3 LAN Topology Implementation

A simple LAN topology was created in GNS3 consisting of three VPCS nodes (PC1, PC2, PC3) connected to one built-in Ethernet switch. Static IP addresses were assigned via the VPCS console using the IP command.

Device	IP Address	Subnet Mask	Tool
PC1	192.168.1.1	255.255.255.0	GNS3 VPCS
PC2	192.168.1.2	255.255.255.0	GNS3 VPCS
PC3	192.168.1.3	255.255.255.0	GNS3 VPCS

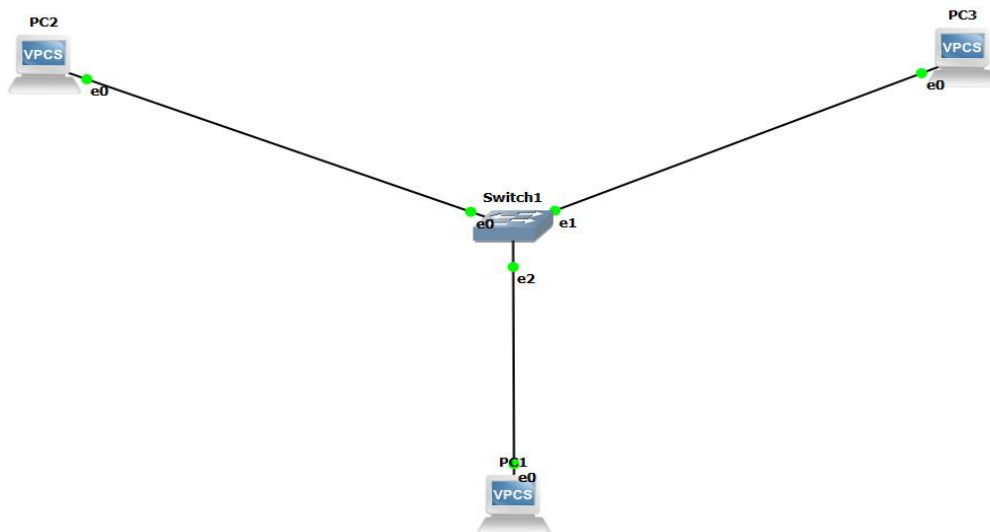


Figure 5.1: GNS3 LAN Topology PC1, PC2, PC3 connected via Ethernet Switch

IP Configuration Commands used in GNS3 VPCS:

- PC1> ip 192.168.1.1 255.255.255.0 then save.
- PC2> ip 192.168.1.2 255.255.255.0 then save.
- PC3> ip 192.168.1.3 255.255.255.0 then save.

6. GNS3 Ping Test Results

After configuring IP addresses on all three VPCS nodes, ping tests were performed to verify end-to-end connectivity. All ping tests were successful with 0% packet loss.

```
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> ping 192.168.1.2
host (192.168.1.2) not reachable

PC1> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.517 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.288 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.374 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.003 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.406 ms

PC1> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=0.863 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=1.154 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.131 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.115 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.129 ms

PC1> █
```

Figure 6.1: GNS3 PC1 pinging PC2 (192.168.1.2) and PC3 (192.168.1.3) Successful

```
ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.098 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.903 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.954 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.467 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=0.924 ms

PC2>
PC2> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.200 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=2.486 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.079 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.333 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.445 ms

PC2> █
```

Figure 6.2: GNS3 PC2 pinging PC1 and PC3 Successful

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.762 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.265 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.958 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.041 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.176 ms

PC3> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.992 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.537 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.959 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.993 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.119 ms

PC3> █
```

Figure 6.3: GNS3 PC3 pinging PC1 and PC2 Successful

7. Cisco Packet Tracer LAN Topology Implementation

The same LAN topology was recreated in Cisco Packet Tracer using three PC-PT end devices and a Cisco 2960-24TT switch. Static IP addresses were assigned through Desktop > IP Configuration panel of each PC.

Device	IP Address	Subnet Mask
PC0	20.31.1.1	255.255.255.0
PC1	20.31.1.2	255.255.255.0
PC2	20.31.1.3	255.255.255.0

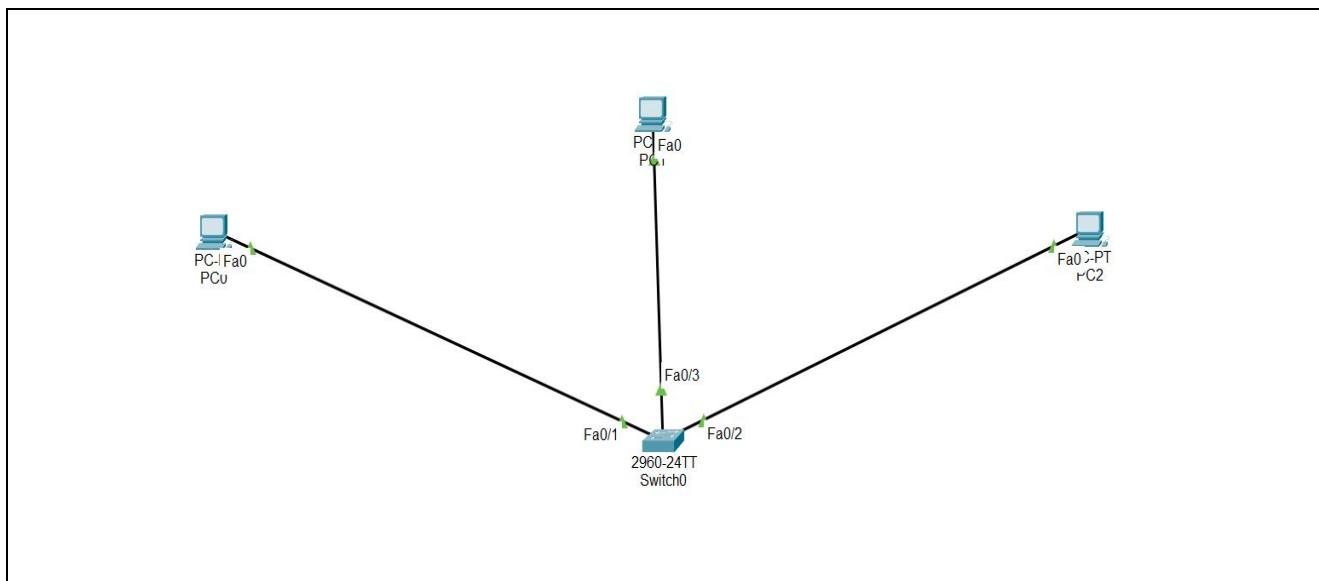


Figure 7.1: Cisco Packet Tracer LAN Topology PC0, PC1, PC2 connected via 2960-24TT Switch

8. Cisco Packet Tracer — Ping Test Results

Ping tests were performed from all three PCs using the Command Prompt in Packet Tracer's Desktop tab. All ping tests returned 4 out of 4 packets with 0% loss, confirming successful LAN connectivity.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 30.20.1.2

Pinging 30.20.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 30.20.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 20.31.1.2

Pinging 20.31.1.2 with 32 bytes of data:

Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time=9ms TTL=128
Reply from 20.31.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 20.31.1.3

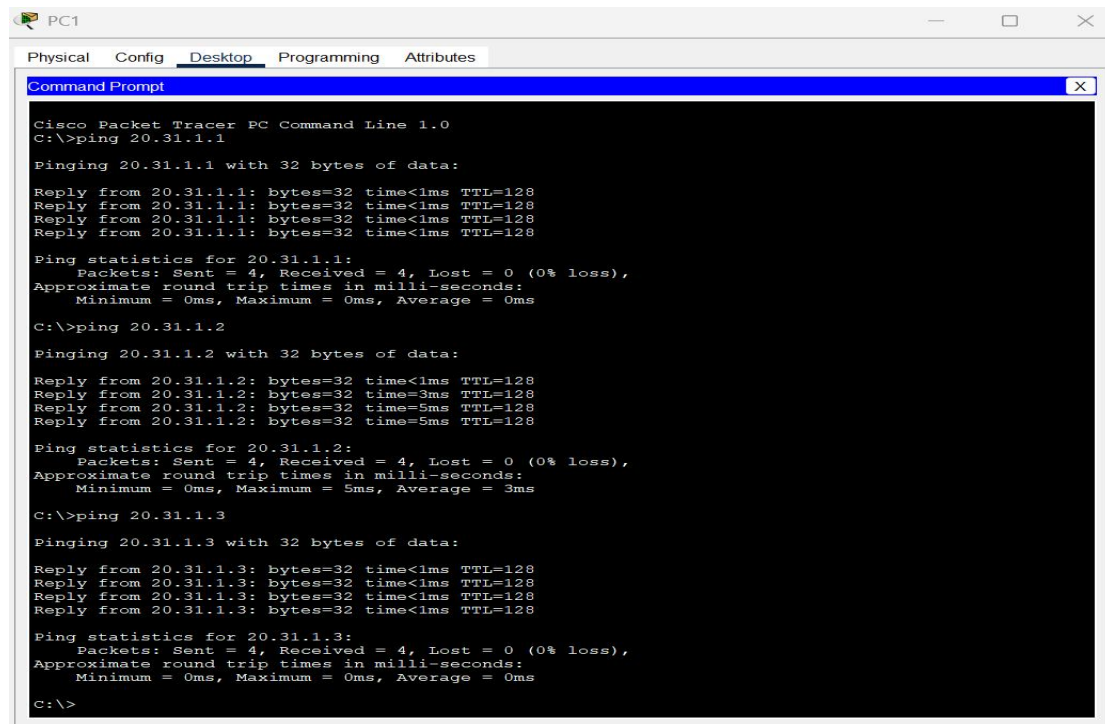
Pinging 20.31.1.3 with 32 bytes of data:

Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 8.1: Packet Tracer PC0 pinging PC1 (20.31.1.2) and PC2 (20.31.1.3) Successful



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.31.1.1

Pinging 20.31.1.1 with 32 bytes of data:

Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 20.31.1.2

Pinging 20.31.1.2 with 32 bytes of data:

Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time=3ms TTL=128
Reply from 20.31.1.2: bytes=32 time=5ms TTL=128
Reply from 20.31.1.2: bytes=32 time=5ms TTL=128

Ping statistics for 20.31.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 3ms

C:\>ping 20.31.1.3

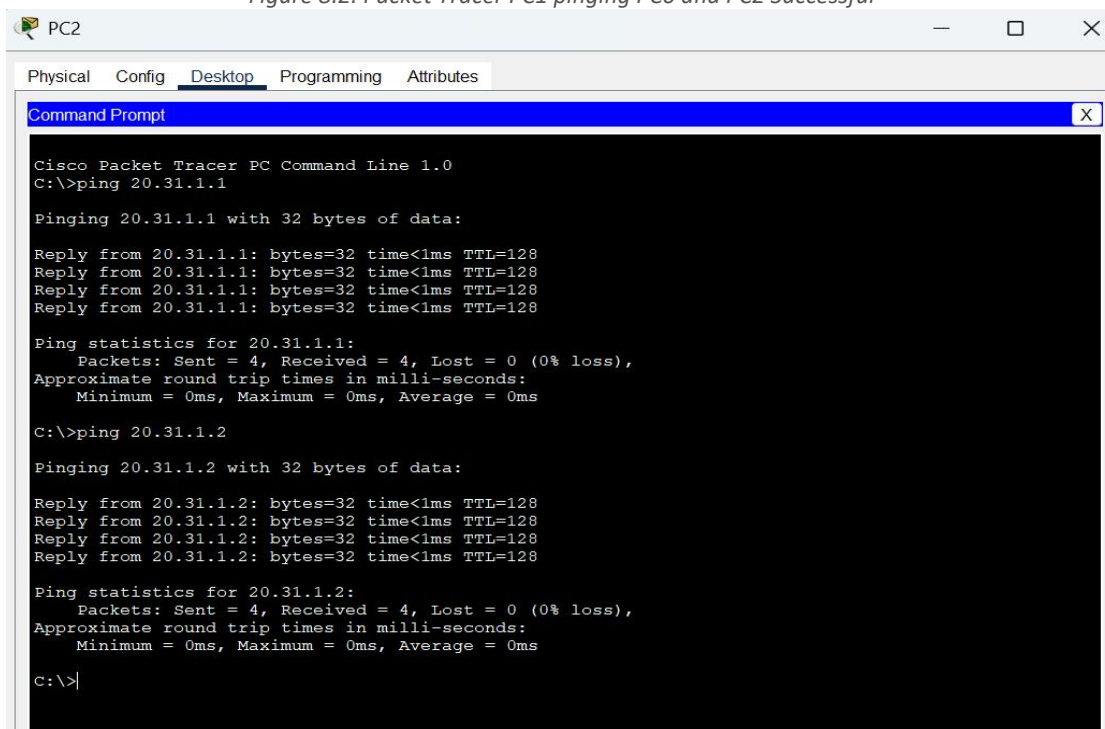
Pinging 20.31.1.3 with 32 bytes of data:

Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128
Reply from 20.31.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 8.2: Packet Tracer PC1 pinging PC0 and PC2 Successful



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.31.1.1

Pinging 20.31.1.1 with 32 bytes of data:

Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128
Reply from 20.31.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 20.31.1.2

Pinging 20.31.1.2 with 32 bytes of data:

Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time<1ms TTL=128
Reply from 20.31.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 20.31.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 8.3: Packet Tracer PC2 pinging PC0 and PC1 Successful

9. Challenges Faced

GNS3 IOS Image Setup

GNS3 does not include router/switch images by default. Cisco 3745 and 7200 IOS images had to be manually imported via Edit > Preferences > Dynamips > IOS Routers. The Idle-PC value was also configured to prevent excessive CPU usage.

HAXM Installation Error

During GNS3 installation, Intel HAXM failed to install due to system incompatibility. This was resolved by skipping HAXM as it is only required for Android emulation and is not needed for Cisco IOS-based labs.

Wrong IP Address Entry

During initial VPCS configuration in GNS3, an incorrect IP was entered for PC2. This was corrected by re-entering the correct IP command, which automatically overwrites the previous value.

10. Conclusion

This week's assignment provided a strong foundation in computer networking concepts and practical hands-on experience with two industry-standard simulation tools. Both Cisco Packet Tracer and GNS3 were successfully installed, configured, and used to build a simple LAN topology.

Static IP addresses were assigned to all end devices, and end-to-end connectivity was verified through successful ping tests in both environments. The key difference I observed is that Packet Tracer is more beginner-friendly with a graphical IP configuration interface, while GNS3 provides a more realistic command-line environment using actual Cisco IOS images.

By the end of this week, the following learning outcomes were achieved: understanding of LAN fundamentals, IP addressing and subnetting, proficiency in navigating both simulation tools, ability to configure and verify network connectivity, and preparation of professional technical documentation.